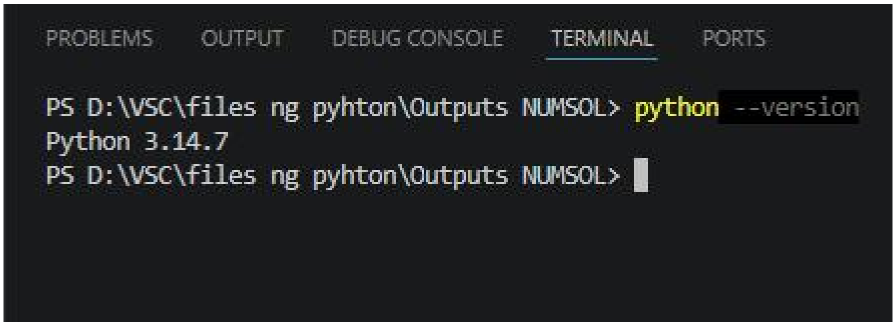
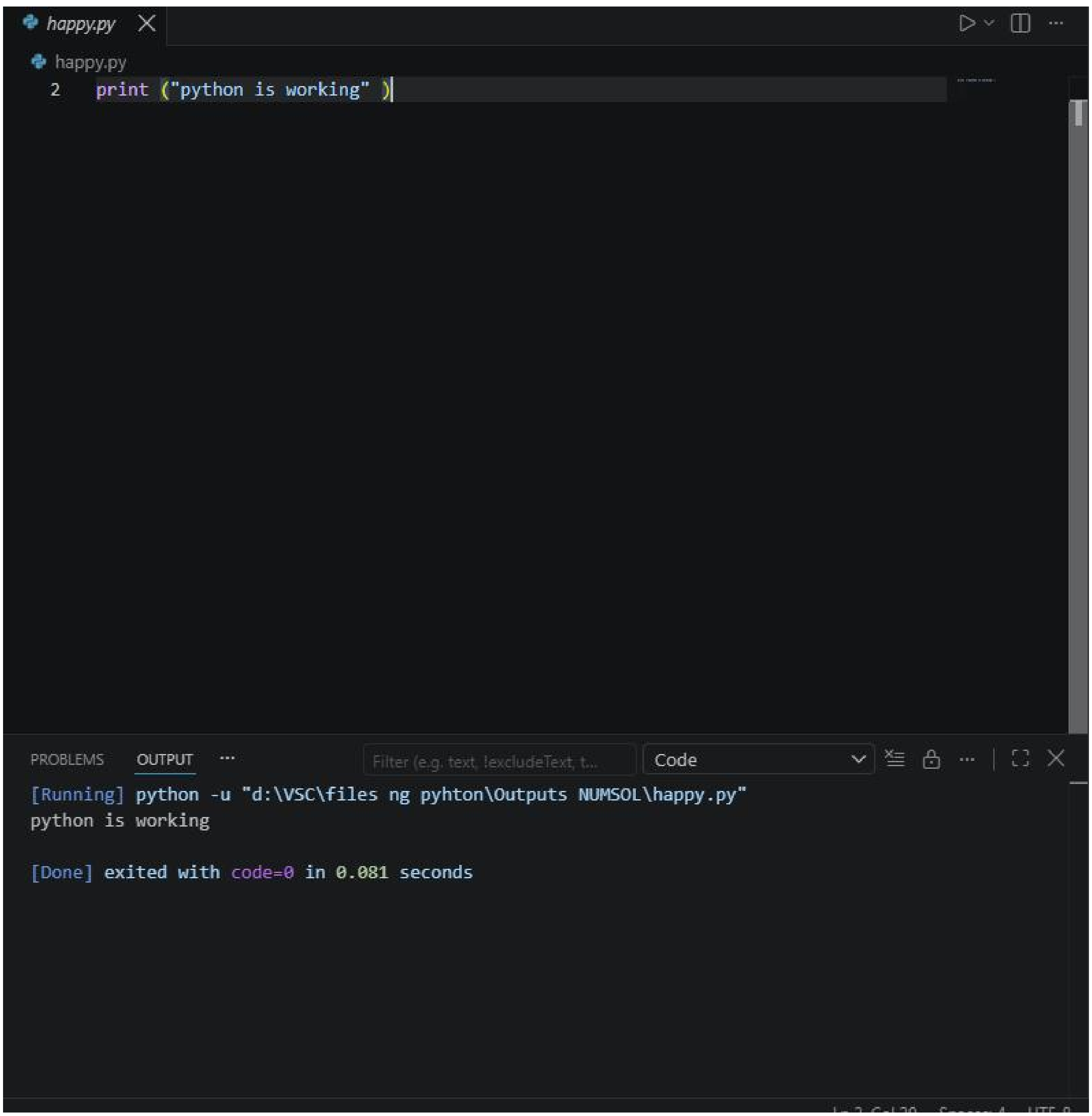
1. Python Version – 3.14.7



2. Python working



3 ><4. S—▱▱○ⓈⓈ//—Ⓛ ∠∅⌓ PⓁ∠⇗//□◎Ⓜ ◎○∠⌓▷.

lab00_check_Balobal.py

```python
# ---- 3. SciPy cross-check ----
area, _ = integrate.quad(np.sin, 0, np.pi)
print("integral of sin(x) from 0 to pi :", round(area, 12), "(exact value 2)")

# ---- 4. Matplotlib figure ----
os.makedirs("figures", exist_ok=True)
plt.rcParams.update({'figure.dpi': 150})

fig, ax = plt.subplots(figsize=(5.2, 3.2))
ax.plot(x, y, color="#b05a5a", lw=1.6, label="sin(x)")
ax.plot(x, np.cos(x), color="#85a800", lw=1.2, ls="--", label="cos(x)")
ax.set_xlabel("x (radians)")
ax.set_ylabel("value")
ax.set_title("Lab Assignment 0 - platform check")
ax.grid(True, alpha=0.3)
ax.legend()
fig.tight_layout()
fig.savefig("figures/lab00_check.png")

print("\nfigure written to figures/lab00_check.png")
print("PLATFORM READY.")
```

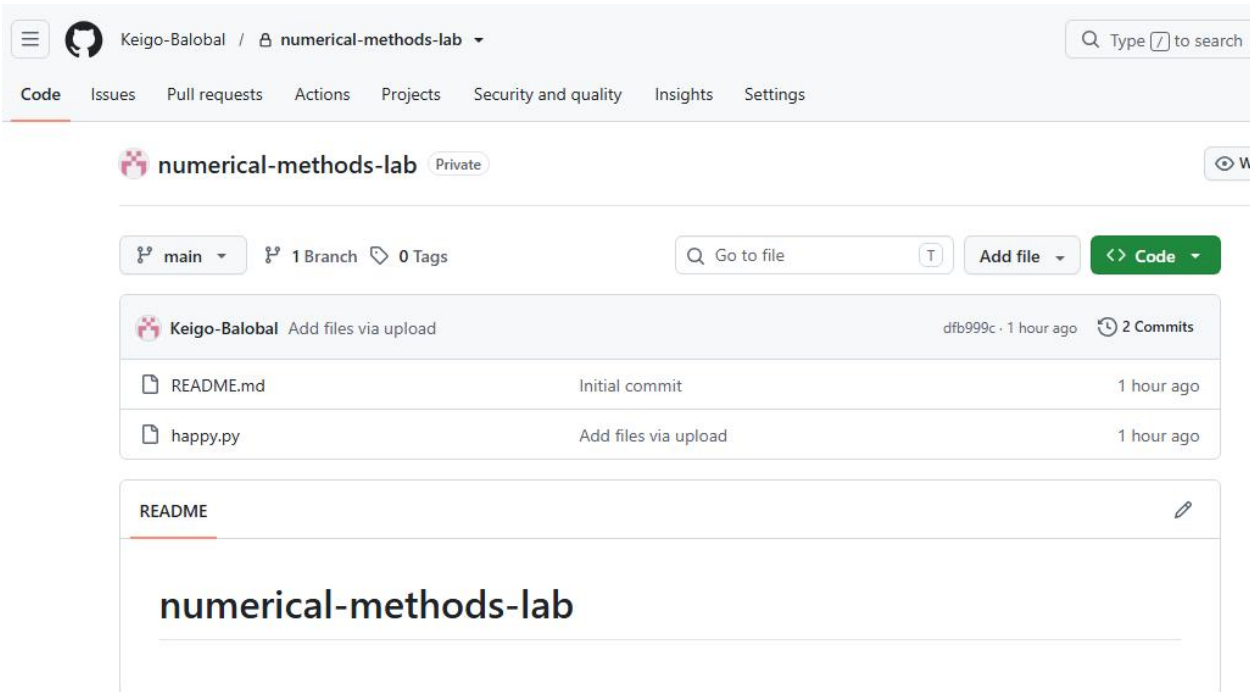
PROBLEMS OUTPUT

```
[Running] python -u "d:\VSC\files ng pyhton\Outputs NUMSOL\lab00_check_Balobal.py"
Python  3.14.7
NumPy  2.5.2
SciPy  1.18.0
Matplotlib  3.11.1

mean of sin(x) over one period : 0.0
integral of sin(x) from 0 to pi : 2.0 (exact value 2)

figure written to figures/lab00_check.png
PLATFORM READY.
```

5.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Code
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2 Commits
README.md
Initial commit
1 hour ago
happy.py
Add files via upload
1 hour ago
README
numerical-methods-lab